

## SUBHOJYOTI MUKHERJEE

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Website: <https://subhojyoti.github.io/>

**Research Interests** Reinforcement Learning ([RL](#)), Large Language Models ([LLM](#)), Reinforcement Learning with Human Feedback ([RLHF](#)), Incontext Learning ([ICL](#)), Optimal Design ([OD](#)).

**Work Experience** **Research Scientist/Engineer, Adobe Inc.**, San Jose, USA 2025, March – present

**Products**

1. Pre-training of a small on-device English language model for Document Overview in Acrobat Reader (launched Sept. 2025); patent filed.
2. RL post-training of a small on-device multilingual language model for Spanish Document Overview in Acrobat Reader (planned 2026).
3. Synthetic data generation and post-training for the Adobe Express Agent Orchestrator (ongoing); patent filed.
4. RL post-training for the Firefly image editing model (ongoing).

**Education** **University of Wisconsin-Madison**, Madison, USA Fall 2019 – 2025, Feb  
*Ph.D.*, Electrical & Computer Engineering  
Adviser: Dr. Robert Nowak, Dr. Josiah Hanna and Dr. Qiaomin Xie

**University of Wisconsin-Madison**, Madison, USA Fall 2019 – 2021  
*M.S.*, Electrical Engineering  
Adviser: Dr. Robert Nowak

**Indian Institute of Technology Madras**, India 2015–2018  
*M.S (Research)*, Computer Science  
Advisers: Dr. Balaraman Ravindran and Dr. Nandan Sudarsanam

**West Bengal University of Technology**, Kolkata, India 2009–2013  
*Bachelor of Technology*, Computer Science & Engineering

**Publications**

1. Yash Savani, Branislav Kveton, Yuchen Liu, Yilin Wang, Jing Shi, **Subhojyoti Mukherjee**, Nikos Vlassis, Krishna Kumar Singh, *Stepwise Credit Assignment for GRPO on Flow-Matching Models*. **CVPR 2026, main conference**
2. **Subhojyoti Mukherjee**, Viet Dac Lai, et.al. *"Offline RL by Reward-Weighted Fine-Tuning for Conversation Optimization"*. **NeurIPS 2025, main conference** [Paper] ([LLM](#), [RL](#))
3. **Subhojyoti Mukherjee\***, Yu Xia\*, et.al. *"From Selection to Generation: A Survey of LLM-based Active Learning"*. **ACL 2025, main conference** [Paper] ([LLM](#))
4. Gaurush Hiranandani, Haolun Wu, **Subhojyoti Mukherjee**, Sanmi Koyejo *"Logits are All We Need to Adapt Closed Models"*. **ICML 2025, main conference** [Paper] ([LLM](#), [RL](#))

5. **Subhojyoti Mukherjee**, Josiah Hanna, Qiaomin Xie, Robert Nowak, "*Pretraining Decision Transformers with Reward Prediction for In-Context Multi-task Structured Bandit Learning*". **RLC 2025** [Paper] ([LLM](#), [ICL](#))
6. **Subhojyoti Mukherjee**, Qiaomin Xie, Robert Nowak, "*Multi-task Representation Learning for Fixed Budget Pure-Exploration in Linear and Bilinear Bandits*". **RLC 2025** [Paper] ([RL](#))
7. **Subhojyoti Mukherjee**, Anusha Lalitha, Kousha Kalantari, Aniket Anand Deshmukh, Ge Liu, Yifei Ma, Branislav Kveton, "*Optimal Design for Human Preference Elicitation*". (**NeurIPS 2024, main conference**) [Paper] ([LLM](#), [RLHF](#))
8. **Subhojyoti Mukherjee**, Josiah Hanna, Robert Nowak, "*SaVeR: Optimal Data Collection Strategy for Safe Policy Evaluation in Tabular MDP*". (**ICML 2024, main conference**) [Paper] ([RL](#))
9. **Subhojyoti Mukherjee**, Anusha Lalitha, Kousha Kalantari, Aniket Anand Deshmukh, Ge Liu, Yifei Ma, Branislav Kveton, "*Optimal Design for K-Way Human Feedback*". (**Models of Human Feedback for AI Alignment workshop ICML 2024**) [Paper] ([LLM](#), [RLHF](#), [OD](#))
10. Aniruddha Bhargava, Lalit Jain, Branislav Kveton, Ge Liu, **Subhojyoti Mukherjee**, "*Off-Policy Evaluation from Logged Human Feedback*". (**Models of Human Feedback for AI Alignment workshop ICML 2024**) [Paper] ([LLM](#))
11. **Subhojyoti Mukherjee**, Qiaomin Xie, Josiah Hanna, Robert Nowak, "*SPEED: Optimal Experimental Design for Policy Evaluation in Linear Heteroscedastic Bandits*". (**AISTATS 2024**) [Paper] ([RL](#), [OD](#))
12. **Subhojyoti Mukherjee**, Qiaomin Xie, Josiah Hanna, Robert Nowak, "*Multi-task Representation Learning for Pure Exploration in Bilinear Bandits*", Neural Information Processing Systems. (**NeurIPS 2023**) [Paper] ([RL](#), [OD](#))
13. **Subhojyoti Mukherjee**, Josiah Hanna, Robert Nowak, "*ReVar: Strengthening Policy Evaluation via Reduced Variance Sampling*". Uncertainty in Artificial Intelligence. (**UAI-22**) [Paper] ([RL](#))
14. **Subhojyoti Mukherjee**, "*Safety Aware Changepoint Detection for Piecewise i.i.d. Bandits*". Uncertainty in Artificial Intelligence (**UAI-22**). [Paper] ([RL](#))
15. **Subhojyoti Mukherjee\***, Ardhendu Tripathy\*, Robert Nowak, "*Chernoff Sampling for Active Testing and Extension to Active Regression*". The 25th International Conference on Artificial Intelligence and Statistics (**AISTATS-22**). [Paper] ([RL](#), [OD](#))
16. Blake Mason, Romain Camilleri, **Subhojyoti Mukherjee**, Kevin Jamieson, Robert Nowak, Lalit Jain, "*Nearly Optimal Algorithms for Level Set Estimation*". The 25th International Conference on Artificial Intelligence and Statistics (**AISTATS-22**). [Paper] ([RL](#), [OD](#))
17. Samarth Gupta, Shreyas Chaudhari, **Subhojyoti Mukherjee**, Gauri Joshi, Osman Yagan, "A Unified Approach to Translate Classical Bandit Algorithms to the Structured Bandit Setting", *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP-21)*. [Paper] ([RL](#))
18. Samarth Gupta, Shreyas Chaudhari, **Subhojyoti Mukherjee**, Gauri Joshi, Osman Yagan, "A Unified Approach to Translate Classical Bandit Algorithms to the Structured Bandit Setting", *IEEE Journal on Selected Areas in Information Theory* (**2020**). [Paper] ([RL](#))

19. **Subhojyoti Mukherjee**, and Odalric-Ambrym-Maillard, “*Distribution-dependent and Time-uniform Bounds for Piecewise i.i.d Bandits*”, *Thirty-sixth International Conference on Machine Learning (ICML-19)*, Workshop on Reinforcement Learning for Real Life 2019 track [Poster]. [Paper] ([RL](#))
20. **Subhojyoti Mukherjee**, K.P. Naveen, Nandan Sudarsanam, and Balaraman Ravindran, “*Efficient UCBV: An Almost Optimal Algorithm using Variance Estimates*”, *Proceedings of the Thirty-Second Association for the Advancement of Artificial Intelligence (AAAI-18)*, main conference track [Oral]. [Paper] ([RL](#))
21. **Subhojyoti Mukherjee**, K.P. Naveen, Nandan Sudarsanam, and Balaraman Ravindran, “*Thresholding Bandits with Augmented UCB*”, *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence (IJCAI-17)*, main conference track [Poster]. [Paper] ([RL](#))

## Preprints

1. Daeun Lee, **Subhojyoti Mukherjee**, Branislav Kveton, Ryan A Rossi, Viet Dac Lai, Seunghyun Yoon, Trung Bui, Franck Dernoncourt, Mohit Bansal, [Paper]
2. **Subhojyoti Mukherjee**, Stefano Petrangeli, Branislav Kveton, Trung Bui, Franck Dernoncourt, Arko Mukherjee, “*Agentic Planning with Reasoning for Image Styling via Offline RL*”. [Paper] ([in submission to ICML 2026](#))
3. Puneet Mathur, Branislav Kveton, **Subhojyoti Mukherjee**, Viet Dac Lai, “*Partial Policy Gradients for RL in LLMs*”. [Paper] ([in submission to ICML 2026](#))
4. **Subhojyoti Mukherjee**, Anusha Lalitha, Sailik Sengupta, Aniket Anand Deshmukh, Branislav Kveton, “*Multi-Objective Alignment of Large Language Models Through Hypervolume Maximization*”. [Paper] ([LLM](#), [RLHF](#))
5. **Subhojyoti Mukherjee**, Ge Liu, Aniket Anand Deshmukh, Anusha Lalitha, Yifei Ma, Branislav Kveton, “*Optimal Design for Adaptive In-Context Prompt Tuning in Large Language Models*”. [Paper] ([LLM](#), [ICL](#), [OD](#))
6. **Subhojyoti Mukherjee**, Ruihao Zhu, Branislav Kveton, “*Efficient and Interpretable Bandit Algorithms*”. [Paper] ([RL](#), [OD](#))

## Mentorship:

**Primary Mentor:** Daeun Lee — StreamGaze project (paper submitted to CVPR 2026)

**Co-Mentor:** Junghyun Lee; Yash Savani; Chima Adiole; Reuben A. Luera; Mohammed Elmmoghany; Bo Ni

## Research Internships

1. **Amazon.com Inc., Santa Clara, USA:** Summer 2024 (Full-time), Host: Branislav Kveton, Yifei Ma, Anusha Lalitha, Kousha Kalantiri, Aniket Deshmukh. Working on Alignment for Multi-objective optimization with LLMs.
2. **Amazon.com Inc, Santa Clara, USA:** Fall 2023 (Part-time), Host: Branislav Kveton, Yifei Ma, Anusha Lalitha, Kousha Kalantiri, Ge Liu, Aniket Deshmukh, Anoop Deoras. Working on RLHF with LLMs
3. **Amazon.com Inc, Santa Clara, USA:** Summer 2023 (Full-time), Host: Branislav Kveton, Yifei Ma, Anusha Lalitha, Ge Liu, Aniket Deshmukh, Anoop Deoras. Worked on Active In-Context Learning with LLMs
4. **CMU, ECE Dept., Pittsburgh, USA:** Summer 2019, Host: Gauri Joshi. Worked on Structured Bandits

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|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p>5. <b>Adobe Research, San Jose, USA:</b> Spring 2018. Host: Branislav Kveton. Worked on Item recommendation with Ranking and Bandits</p> <p>6. <b>INRIA, SequeL Lab, Lille, France:</b> Fall 2017, Host: Odalric Maillard. Worked on Non-stationary Bandits</p>                                                                                                                                                                                                                                                                                                                                |
| <b>PhD Thesis (ECE, UW-Madison)</b>     | Adaptive Data Collection for Policy Evaluation, Multi-Task Learning and LLM Alignment[Thesis] ( <a href="#">RL</a> , <a href="#">OD</a> , <a href="#">LLM</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Master's Thesis (EE, UW-Madison)</b> | Active Sequential Hypothesis Testing with Extension to Active Regression and Multi-armed Bandits [Thesis] ( <a href="#">RL</a> , <a href="#">OD</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Master's Thesis (CS, IIT Madras)</b> | Finite-time Analysis of Frequentist Strategies for Multi-armed Bandits [Thesis]( <a href="#">RL</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Teaching Experience</b>              | <p><b>Teaching/Research Assistant</b>, UW-Madison 2019–2025<br/> <i>Matrix Methods in Machine Learning</i> - Prof. Robert Nowak<br/> <i>Mathematical Foundation in Machine Learning</i> - Prof. Robert Nowak</p> <p><b>Teaching Assistant</b>, UMass Amherst 2018–2019<br/> <i>Natural Language Processing</i> - Prof. Mohit Iyyer<br/> <i>Design of Algorithms</i> - Prof. Daniel Sheldon</p> <p><b>Teaching Assistant</b>, IIT Madras 2015–2018<br/> <i>Introduction to Programming</i> - Prof. Raghavendra Rao B. V.<br/> <i>Reinforcement Learning</i>(twice) - Prof. Balaraman Ravindran</p> |
| <b>Reviewer and Service</b>             | <p>1. AISTATS, UAI, AAAI, ICML, ICLR, NeurIPS, TMLR, KDD, RLC, CVPR</p> <p>2. Main Co-ordinator of SILO seminar at UW-Madison</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Award Grants and Fellowship</b>      | <p>1. Top reviewer award for UAI 2023, Neurips 2023</p> <p>2. Student Scholarship for AAAI 2018, UAI 2022, Neurips 2023</p> <p>3. UW-Madison nominee for Apple PhD fellowship and Two-sigma PhD fellowship. Received UW-Madison Chancellor's Opportunity Fellowship 2019-20, UW-Madison ECE Welcome Award of USD 3000.</p> <p>4. IIT Madras student travel grant of USD 2300, Google travel grant of USD 1700, Microsoft travel grant of USD 1435 (declined).</p>                                                                                                                                 |
| <b>Other Achievements</b>               | <p>Ranked 1150/155190 candidates in Graduate Aptitude Test in Engineering (<b>GATE</b>) 2014.</p> <p>Secured 98.93 percentile in Common Admission Test (<b>CAT</b>) 2014 among 196988 candidates.</p>                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>References</b>                       | Available Upon Request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |